Table 1

Forecasting performance across network attack prediction horizons

Forecasting model	Architecture	H5 (5 min)		H10 (10 min)		H15 (15 min)	
		AP	Δ (pp)	AP	Δ (pp)	AP	Δ (pp)
Single LSTM	Recurrent	0.7132	-4.78	0.8963	-0.97	0.9260	-1.00
Stacked LSTM	Recurrent	**0.7610**	+0.00	**0.9060**	+0.00	0.9359	-0.02
XGBoost V2	Boosted trees	0.6763	-8.47	0.8751	-3.09	N/A	
CR-LSTM	Competing risk	0.7248	-3.62	0.8991	-0.69	**0.9361**	+0.00
CR-TCN	Competing risk	0.7174	-4.36	0.8939	-1.21	0.9318	-0.42
Original Graph LSTM	Graph-based	0.7270	-3.40	0.8880	-1.81	0.9240	-1.21
Optimized Graph LSTM	Graph-based	0.7241	-3.69	0.8869	-1.91	0.9275	-0.86
GCN-LSTM	Graph-based	0.7433	-1.78	0.8860	-2.00	0.9190	-1.70
GAT-LSTM	Graph-based	0.7209	-4.01	0.8844	-2.16	0.9181	-1.80

Notes: AP = average precision; Δ = AP difference from the best model at the same prediction horizon, in percentage points.
Bold values mark the highest measured AP per horizon. N/A indicates an unavailable saved model prediction, not a score of zero.
Synthetic NAF v2 development validation (18,839 aligned windows). H5/H10/H15 positives: 1,125 / 2,250 / 3,375.
These descriptive rankings do not establish statistical significance or live-network performance.